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Analysis of a hybrid quantum network for classification tasks

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Abstract

In the era of noisy intermediate scaled quantum computers, one of the possible applications to search for an advantage of quantum computing is machine learning. Here, we report about an analysis, where a hybrid quantum-classical network is applied to classify non-trivial datasets (finance and MNIST data). In comparison to a pure classical network, we find an advantage when looking at several performance measures. As in classical machine learning, problems around overfitting the dataset arise. Therefore, we explore different possibilities to regularise the network.

KEYWORDS

finance, machine learning, MNIST, quantum computing, regularisation, tensorflow

1 | INTRODUCTION

With the availability of quantum computers in the cloud around 2016, a race started towards suitable applications for this technology. While many people explore a potential quantum advantage in the areas of optimisation and simulation, others focus on potential improvements of machine learning methods. Out of this, quantum machine learning (QML) emerged as an approach to combine classical machine learning ideas (e.g. deep neural nets) with the peculiarities of quantum computations–superposition and entanglement. From a technological point of view, we are still in the NISQ-era [1], that is, where only noisy intermediate scaled quantum computers are available. Although the number of qubits and the depth of possible circuits are limited, the promises of QML can be explored nevertheless. This is done with techniques like training via backpropagation, regularisation or learning curves well-known in the domain of machine learning.

In this article, we present a hybrid quantum network ansatz that is designed to classify non-trivial and/or more realistic datasets. 'Realistic' data which appear in industrial or finance use cases tend to have many features, many observations and sometimes the target is unbalanced–one target event is more populated than the other(s).

The next section introduces a hybrid classical quantum network to leverage the ability of pure QML. In Section 3, we present details of the software framework and in section IV and V numerical results on different datasets can be found. This is followed in Section 6 by a discussion about regularisation within this approach. A summary and outlook concludes the study in Section 7.

2 | SUPERVISED CLASSIFICATION TASKS WITH QUANTUM COMPUTERS

One large field of interest in the growing field of QML consists of the development of quantum classifiers: These are quantum algorithms (protocols) suitable for binary or multiclass classification tasks, see, for example, [2–4]. For an extensive overview of the underlying methods and restrictions, the reader is referred to [5, 6].

In many cases, the different approaches are demonstrated with very simple benchmark datasets: They have only two or three features and only a few training samples. Additionally, the approaches sometimes use only two or three qubits. Therefore, according to the existing literature, it is not obvious if and how a quantum classifier can be developed which is useful for datasets with

- Up to 1000 features.
- More than 1000 training samples.
- An unbalanced target (e.g. a binary classification where class 0 appears 90% in the training data and class 1 appears 10% in the training data).

Note, in typical industry use cases—for example, in the finance sector—such conditions are usually met [7]. This leads

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to the research question to explore the scalability and deployment of quantum algorithms for such realistic datasets. To achieve this goal, it is mandatory to use a robust framework, which allows going beyond proof-of-concept calculations.

A common approach in QML for NISQ, which is also used in most of the mentioned papers above, is named 'variational (or parametrised) quantum classifier' (VQC or PQC). The basic idea can be described as follows: First, the classical data have to be encoded as quantum states. For this task, a couple of different schemes are proposed in the literature: angle encoding, amplitude encoding, basic encoding etc. The reader is referred to [5, 8] for details. Second, a unitary transformation with a number of learnable parameters is applied on the quantum states. Third, the qubits are measured and the measurement results are post-processed and fed into a classical loss function. Through a feedback loop, the learnable parameters of the unitary transformation are adjusted until the loss function is minimised. While the prescribed method is quite generic, different settings have been proposed and evaluated in the literature: Different encoding schemes, different unitary transformations with different depth of entanglement, different ways of measuring the qubits and finally different post-processing procedures, see, for example, [9–11].

The described generic VQC-ansatz, however, has one serious shortcoming: The number of qubits needed to encode the classical data with the prescribed methods scale with the number of features per sample. That means, if the classical data have, for example, one hundred features, roughly one hundred qubits would be needed. While in [12], an ansatz is developed which circumvents this problem via data re-uploading, in this contribution we propose a modified, hybrid ansatz, where the encoding is partially performed via a classical neural network. The neural network is used to scale down in a trainable way the number of features in the training data to the parameters used in the quantum network. This hybrid network is schematically shown in Figure 1 in the case of three qubits. To leverage this approach, we combine it with the data re-uploading idea of reference [12]. Data re-uploading is based on the concept, that the encoding part of the network is repeated several times, cf. Figure 2. Basically, as it is shown in reference [12], a universal classifier could be built with only one qubit.

Due to the classical networks in front and after the quantum network, the number of qubits can be chosen independently of the number of features and the number of output classes. In the last classical layer, we use a softmax activation function to feed the results of the network into the loss function. From a technical point of view, our proposed hybrid quantum-classical network is a deep neural net with some quantum layers in the middle.

In the most straightforward design, the structure of the encoding classical neural net is fully determined by the number of features and the number of qubits. It is possible to use more and/or hidden layers in the encoding network in a straightforward way. However, this would lead to a more difficult evaluation of the approach. Here, we do not aim to find the 'best' network, but to understand the interplay between the classical and the quantum part, so we stick to the straightforward design.

The quantum part of the network is constructed in the usual way and starts with all qubits in the $|0>$ state. Then to each qubit i the transformation $R_y(\varphi_{i1})R_z(\varphi_{i2})R_y(\varphi_{i3})$ is applied. $R_y(..)$ and $R_z(..)$ denote rotation gates around the y- and the z-axis, respectively. Note, the angles of the rotations, that is the φ_{ij}, are fed by the classical encoding network. The last step of the encoding part is to entangle pairs of qubits via CNOT-gates.

After the encoding part of the circuit, the variational part $\hat{U}(\phi_1, .., \phi_n)$ is applied. In the literature a couple of different approaches have been suggested, for example, [3, 8, 11].

In this work, we use parametrisation, where to each qubit a combination of $R_y(..) \cdot R_z(..) \cdot R_y(..)$ and CNOT-gates to entangle pairs of qubits are applied. In Figure 2, we show this architecture for the case of three qubits and one layer.

The number of parameters n of the variational part is given by $n = 3 \cdot n_{qubits} \cdot n_{layers}$: For each qubit there are three parameters and the rotation-entanglement setup can be repeated n_{layers} times. We will distinguish two different approaches: In the first approach, the data is loaded once into the network and then the

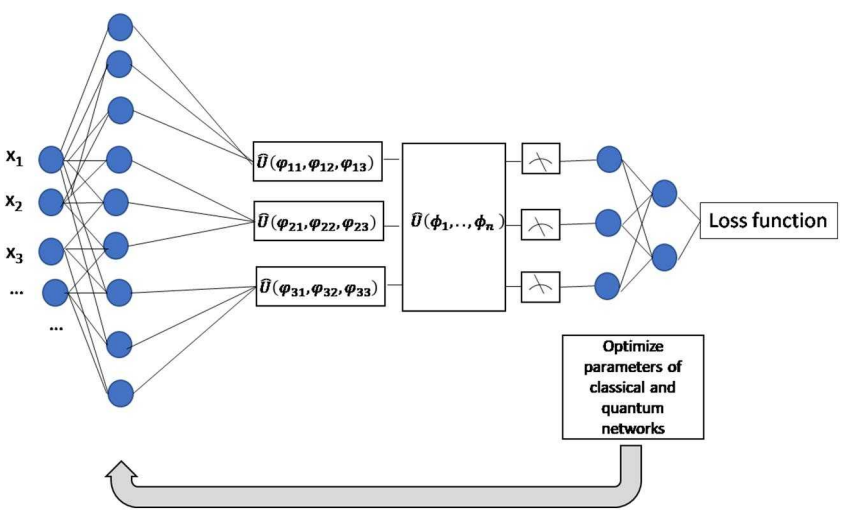

FIGURE 1 Hybrid classical quantum network

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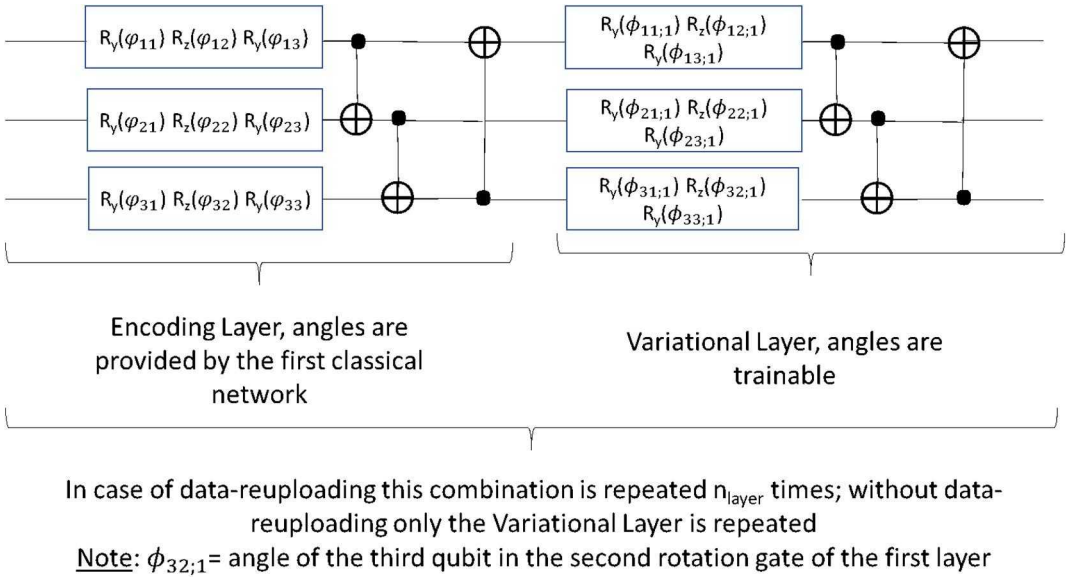

FIGURE 2 Layout of the quantum network

variational part is applied n_{layers} times. In the second approach, we implement the data re-uploading idea, suggested in reference [12], that is together with the variational part, the data encoding circuit is repeated n_{layers} times, too, see Figure 2.

In [13] a comparable but yet a different hybrid learning approach has been proposed: In the spirit of transfer learning in a full classical deep learning model only the last layer is substituted with a quantum network. More in-line with our approach are the results in [14]: There, however, the authors implement no entanglement between the qubits. However, generally it is expected that entanglement of qubits will improve the quality of QML. Furthermore, the authors do not use a second classical network to post-process the measurements, which leads to a lack of flexibility.

To evaluate the ability of our approach, denoted as 'QNet' in the following, we compare the results with a fully classical neural net (denoted as 'NNet') where the quantum network is substituted with one hidden layer. The number of neurons of the hidden layer are tuned to have an approximately equal number of parameters in both networks.

3 | CHOICE OF THE SOFTWARE STACK

Popular frameworks like Qiskit [15] or PennyLane [16] allow building the QML algorithms in a straightforward way. However, in this work, we selected Google's TensorflowQuantum [17] for the following reasons: TensorflowQuantum is embedded in Google's AI ecosystem and allows a seamless training procedure of classical and quantum networks. Due to highly optimised interfaces to the underlying C++-libraries its performance allows to deal with the large datasets of our scope. Using TensorflowQuantum even large quantum circuits (e.g. 10 qubits and 10 layers) can be simulated on a notebook with 16 GByte memory and a Pentium i7 processor within a few hours. All reported results have been obtained with the ADAM-optimiser in tensorflow and a learning rate of 0.01. As loss function, we use categorical crossentropy.

Yet, up until now, Google provides only restrictive access to its quantum hardware; it is therefore at the moment hardly possible to leave the world of the quantum simulators. However, the findings obtained with TensorflowQuantum can be translated to other frameworks like Qiskit in cases where the deployment feature is less important. Implementing this approach in different software frameworks and using different quantum hardware will be one of the next steps in this project.

4 | RESULTS FOR A FINANCE DATASET

4.1 | Balanced dataset–Sample I

As a first use case, we show the results on a credit scoring dataset [18] with six features, scaled to the range of [0,1]. While in the full dataset the default rate is about 16.67%, we prepared a sample (in the following sample [I]) where the default rate was lifted to 50% in order to get a balanced dataset. The training data then consists of 340 observations. In a second sample (in the following sample [II]), we stick to the original unbalanced situation and work with 900 training cases.

In order to compare different classification algorithms, it is important to use appropriate measures and not stick to one particular measure. Especially the commonly used accuracy (ACC) measure can be misleading in the case of unbalanced datasets. Additionally, in general, the ACC depends on the cutoff value between the classes. Therefore, for binary classification we use the area under curve (AUC) as a second performance measure. As it is a common practice in machine

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learning, we not only report on results on the training set but also on separated test data.

In Table 1, we report our results for a calculation with $n_{qubits} = n_{layers} = 8$.

More important than the absolute value of the reported numbers are the relations between them. Lifting the absolute value would always be possible by tweaking the hyper-parameters (e.g. activation functions etc.) of the networks.

For this particular parameter set, some generic features appear which we also observed for other parameters (n_{qubits} and n_{layers}) and hyper-parameter sets:

- In comparison with a classical neural net with approximately the same number of parameters, the hybrid approach shows a better performance.
- Using data re-uploading is superior to an approach where only the variational part is repeated n_{layers} times.
- As in classical machine learning, overfitting can be observed when a trained network is applied to data not seen during training.

4.2 | Unbalanced dataset–Sample II

In Table 2, we report the results for sample II. To make the comparison easier, we stick to the same parameters, that is $n_{qubits} = n_{layers} = 8$.

In this dataset, an ACC of 0.83 would be obtained just by assigning all data points to the major class (the 'non-defaulted' customers). For this reason, it is necessary to consider a second metric to evaluate different models: Although, the approaches with and without data re-uploading seem to be equal when considering just ACC, the values for AUC are different.

As in sample I, it can be seen that overfitting the data is an issue, so one must think about proper regularisation to keep the algorithm flexible enough to deal with unseen data.

TABLE 1 Results of sample I

Network	ACC_{train}	AUC_{train}	ACC_{test}	AUC_{test}
QNet without data re-uploding	0.68	0.74	0.59	0.65
QNet with data re-uploding	0.78	0.86	0.68	0.72
Classical NNet	0.62	0.66	0.62	0.77

Abbreviations: ACC, accuracy; AUC, area under curve.

TABLE 2 Results of sample II

Network	ACC_{train}	AUC_{train}	ACC_{test}	AUC_{test}
Quantum without data re-uploding	0.84	0.72	0.83	0.55
Quantum with data re-uploding	0.84	0.77	0.84	0.55
Classical NNet	0.83	0.66	0.83	0.51

Abbreviations: ACC, accuracy; AUC, area under curve.

5 | RESULTS FOR THE MNIST DATASET

To test our approach for another non-trivial dataset, we use the benchmark dataset MNIST [19]. It consists of a training set of 60,000 grey-scaled pictures of size 28 × 28 where each contains a hand-written number from 0 to 9. Additionally, there is a test set of 10,000 pictures.

In a brute force approach, handling 28 × 28 = 784 features with QML would be hardly possible since it requires 784 qubits to encode the data, which is at the moment out of scope for NISQ devices. However, using a classical neural network in front of the quantum circuit allows downscaling the data to the number of available qubits. Of course, it is expected that the quality of the classification scales with the number of the qubits, so more available qubits would be better for the classification measures.

Classifying hand-written numbers for 0 to 9 is a multiclass classification problem with 10 labels. Solving such a problem from scratch would require to decide between a 'one-versus-one' and a 'one-versus-all' strategy. Within the tensorflow framework, however, switching between binary and multiclass classification is straightforward.

In our study, we choose a training set of 1000 pictures and use the full out-of-sample test set of 10,000 pictures for evaluation. In order to judge the performance of the approach we plug in the same data into a classical network with the same number of parameters as described above.

As a first result, we show in Figure 3 the comparison of the training process with the QNet and the NNet. It can be seen that the learning progress is much faster with the QNet. The same behaviour is observed when comparing the results on the independent test data not seen in the training phase. The hybrid quantum-classical net is superior in detecting the structure of the data.

As a second result, we report the confusion matrix of the QNet for the training data, Figure 4. The confusion matrix is almost 'perfect', showing very few data points out of the diagonal. Moreover, no systematical mis-classification is detected.

When comparing the ACC of the training set (ACC = 0.98) with the ACC of the test set (ACC = 0.81), overfitting is detected. As in the case of the finance data the performance with the NNet (ACC = 0.83 train and ACC = 0.81 test) is slightly worse and shows less overfitting.

Another result of our study are dependencies of the results, ACC of the training and testing set, respectively, on the number of qubits and number of layers. In Figure 5 it is shown for the MNIST data, how the training data's ACC depends on the number of layers; with the number of qubits in [2, 4, 8, 12]. With growing n_{qubits} the dependency on the number of layers is getting weaker. Additionally, the difference between the ACC calculated with 8 and 12 qubits is small, independent on the number of layers. Additionally, Figure 6 supports the observation that a certain complexity of the hybrid net (loosely described by the number of qubits and the number of layers used in the calculation) is needed: Here, we present the dependency of the training ACC on n_{qubits} for different values of

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n_{layers}. Between two and four qubits there is a jump in the ACC. Then a plateau is reached and the results hardly depend on n_{layers} and n_{qubits} anymore. However, the results presented in Figures 5 and 6 are obtained with data re-uploading, and we checked that qualitatively the same results can be obtained without data re-uploading. The same statement is valid for the ACC of the test set.

We conclude, that classification of a certain dataset (here: MNIST data) requires a certain complexity of the hybrid net which can be achieved either by using more quits or more layers. After the gross features of the datasets are described, only small gains in ACC can be expected. Whether this ACC gain is needed depends on the use case where the model is implemented.

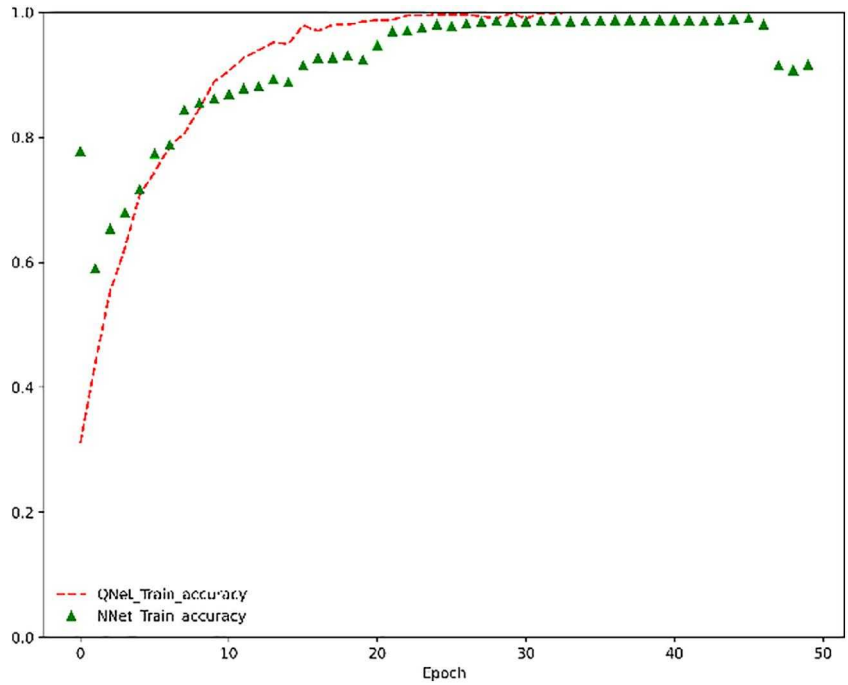

FIGURE 3 Traning of the QNet compared to the NNet

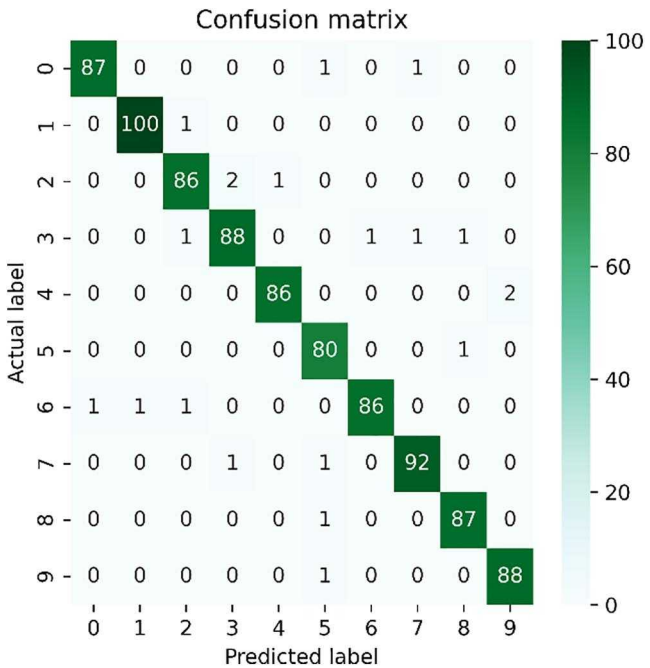

FIGURE 4 QNet's confusion matrix on the training data

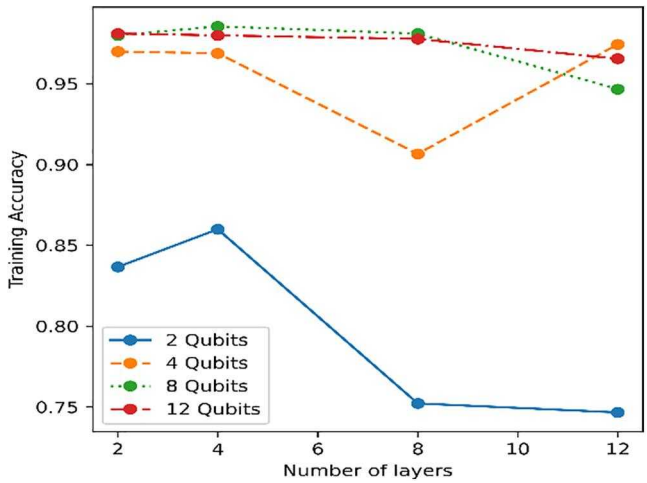

FIGURE 5 Training accuracy of MNIST data as a function of n_{qubits}

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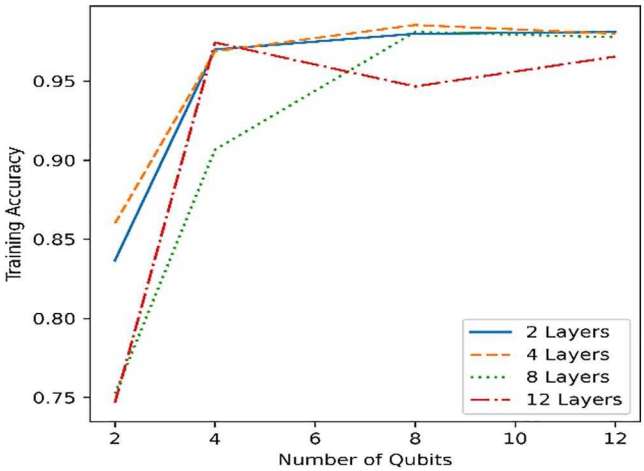

FIGURE 6 Training accuracy of MNIST data as a function of n_{layers}

6 | REGULARISATON OF THE HYBRID NETWORK

In classical machine learning, and especially deep learning, the phenomenon of overfitting is well-known and different recipes have been proposed to heal it [20]. However, in QML the problem of overfitting and therefore proper regularisation schemes are not yet in the focus of research; [21] is one of the few papers dealing with this issue.

To assess this problem in our hybrid approach with the MNIST we tried different possibilities to get comparable results for training and test ACC. Basically, we follow the strategy to distort the parameters of our network in the training phase. The simplest method is to add a dropout layer between the quantum and the second classical net to set randomly the measurement values with an adjustable probability to zero. Fixing this parameter to 0.58 we achieve our result to get training ACC $\approx$ test ACC $\approx$ 0.75 after 30 epochs in the training. Here, we use $n_{layers} = n_{qubits} = 8$ and the data re-uploading setup, see Figure 7. The next step would be to add more training data to raise these values. Interestingly, in the pure classical net we get a value of 0.3 for the training ACC.

Another possibility is to add a regularising term to the loss function of the hybrid network. We use the L2-norm of the angles fed into the quantum network and the L2-norm of the measurement values. Each term is multiplied with a hyper-parameter to adjust the strength of the regularisation individually. We found that the regularisation of the angles fed into the quantum net destroys the ability to classify the data in a sensible way. The regularisation of the measurement results, however, can be fine-tuned in a way to avoid overfitting, see Figure 8. The difference of the training and validation ACC and therefore overfitting is significantly reduced. Using a larger set of training data would presumably lead to an overall higher ACC.

We conclude that regularisation of the proposed hybrid network is possible by regularising the classical measurements of the qubits. Therefore, it is not mandatory to regularise the parameters of the quantum network part which would be more involved.

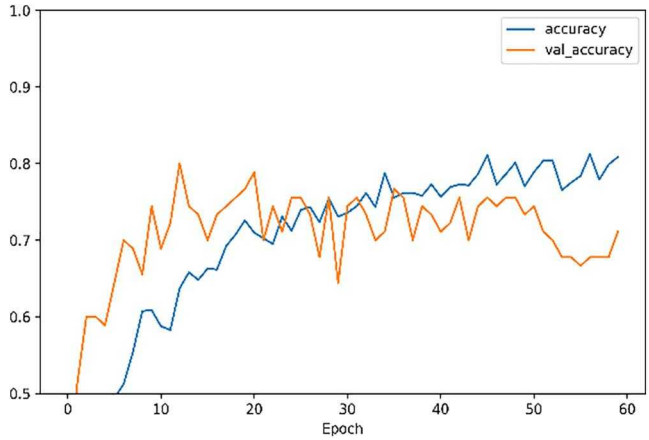

FIGURE 7 Training with Dropout-regularisation of the hybrid network

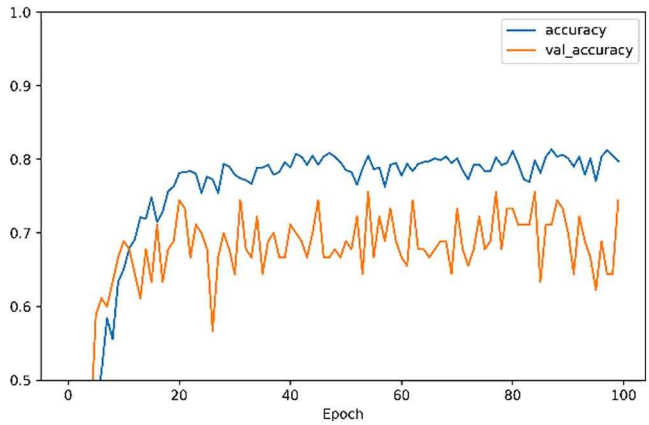

FIGURE 8 Training with L2-regularisation of the hybrid network

7 | SUMMARY AND OUTLOOK

In this study, we introduce a hybrid quantum classical neural net for binary und multiclass classification. In order to allow a seamless interaction of the classical and the quantum part we use the TensorflowQuantum. In this framework, all the well-known tools of classical machine and deep learning are available to train and evaluate the networks. Furthermore, due to the embedding in Google's AI universe it allows in principle deployment even on mobile devices in the near future.

The hybrid quantum classical approach is built on the idea to squeeze a quantum network between two classical networks. The first network transforms the input data to suitable parameters of the quantum part. As it was shown, even high dimensional data as the MNIST dataset, where the number of features is of the order 10^3, can be handled. The classical output network is used for post-processing the measurements of the qubits and is finally connected to an appropriate loss function.

Due to the flexibility of the approach and its implementation with TensorflowQuantum, testing different

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encoding schemes and variational networks is straightforward.

In the case of the finance dataset, we observe that the hybrid network performs slightly better when the classes are balanced. However, even in the unbalanced case, the network is able to distinguish default and non-default cases. When dealing with this dataset, we realised that it is important to use not only ACC as the relevant performance measure, but also AUC. Furthermore, for this dataset the superiority of the data re-uploading approach has been demonstrated.

As a benchmark, we compared the results with a pure classical network, NNet, where the input and output layers are the same and the middle layer has the same number of parameters as the hybrid network QNet. In general, the performance of the QNet as measured in terms of ACC and AUC is in general better than that of the NNet.

For the MNIST dataset, we were able to show that the convergence of the hybrid net to a certain level of ACC is faster than in the classical case. Furthermore, the confusion matrix shows that the network is able to distinguish between the 10 hand-written numbers.

The comparison of the results for different configurations, comparing n_{layers} and n_{qubits} suggests that a certain 'complexity', that is number of parameters, is necessary to achieve the classification task. When going beyond only a small improvement of, for example, the ACC is obtainable.

The exploration of different possibilities to regularise the hybrid network to avoid overfitting, lead to the conclusion that this task is achievable by regularising parameters of the classical networks.

These results—even though they are interesting for their own—provoke a number of questions which will be addressed as next steps in this project:

- As it is the case in classical machine learning, the suitability of a certain network depends on the specific data. Therefore, it is necessary to distinguish generic features of the hybrid network from data-related features and understand how they are connected.
- The interdependence of the different performance measures for quantum classification has to be examined in more detail.
- Although our approach to regularise the network can be pragmatically used, a more thorough investigation should be performed. Especially if and how it is possible to find a scheme to regularise the quantum parameters.
- While all calculations so far have been done with a quantum simulator and therefore an ideal environment, it should be explored how a realistic error rated on physical quantum devices affects the results.

ACKNOWLEDGEMENT

Valuable discussions and support of M. Broughton (Google Research) about implementing the hybrid approach are gratefully acknowledged.

CONFLICT OF INTEREST

No.

PERMISSION TO REPRODUCE MATERIALS FROM OTHER SOURCES

None.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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